D4.3 Climate change

Key words:
Phenology
Photoperiod
Spruce beetle

D4 Continuity and change
Ecosystems

HL

Understandings:

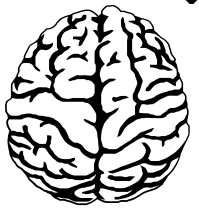
- D4.3.9—Phenology as research into the timing of biological events
- D4.3.10—Disruption to the synchrony of phenological events by climate change
- D4.3.11—Increases to the number of insect life cycles within a year due to climate change
- D4.3.12—Evolution as a consequence of climate change

IB Learner Profile

Inquirers - We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.



Knowledgeable - We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.



Thinkers - We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions



Communicators - We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.



Principled - We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.



Open minded - We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.



Caring - We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.



Risk takers - We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.



Balanced - We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.



Reflective - We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.



D4.3.9—Phenology as research into the timing of biological events

Phenology - the study of periodic events in biological life cycles.



How do plants know when to create new leaves or to flower?

How do birds know when to migrate or to nest?

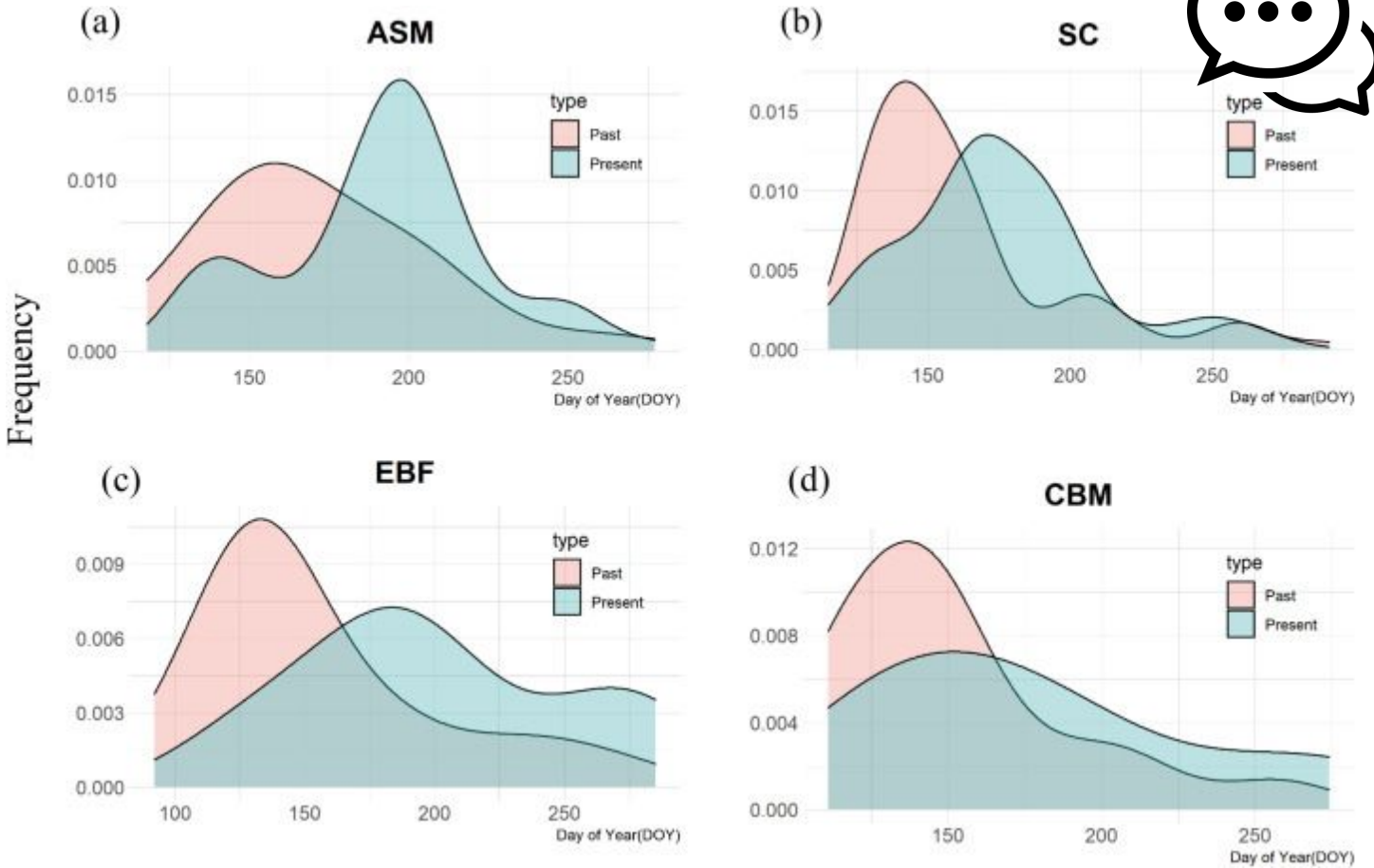
How do animals know when to reproduce?

Photoperiod = Day length (or night length)

Temperature

Rainfall

Discuss this graph



Past = 1950 - 1988

Present = 1989 -2018

'Frequency' represent the probability of flowering times

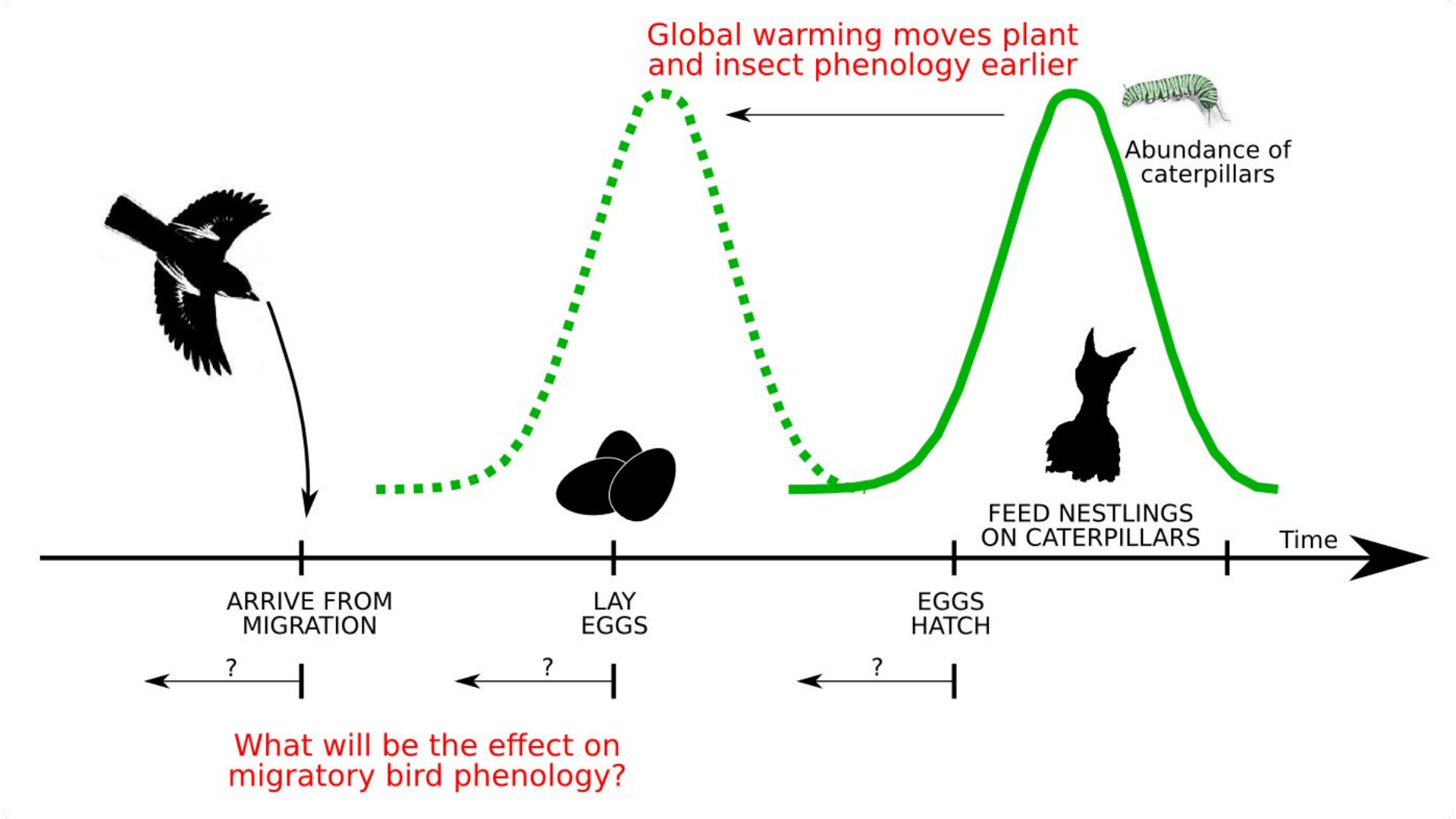
D4.3.10—Disruption to the synchrony of phenological events by climate change



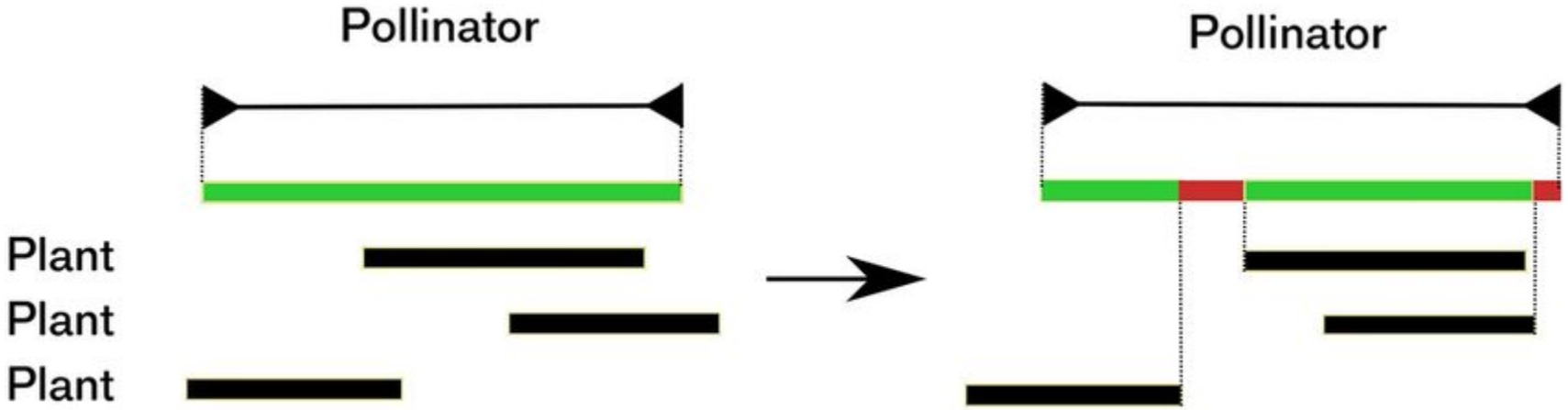
As climate changes phenological events also change.

Leaf unfolding and flowering is advancing by approximately 5–6 days per °C warming

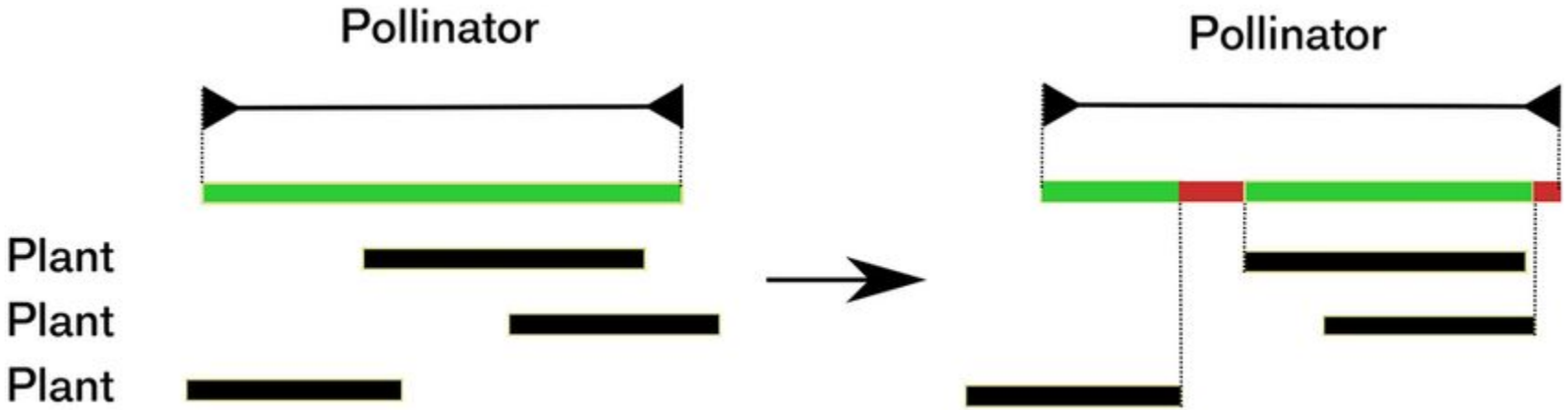
Is this necessarily a bad thing?


Global warming moves plant and insect phenology earlier
Abundance of caterpillars
FEED NESTLINGS ON CATERPILLARS
Time
ARRIVE FROM MIGRATION
LAY EGGS
EGGS HATCH
?
?
?
What will be the effect on migratory bird phenology?

Discuss this graph



Discuss this graph



Phenological differences between plants and pollinators after a change of climate (adapted from Memmott et al. (2007)). Left, the situation before the change. The pollinator is viable, because there are plants that flower during all its activity period. Right, situation after climate change. The activity periods changed somewhat randomly. The pollinator has two parts of its activity period without a plant to pollinate, which leads to its disappearance in the model

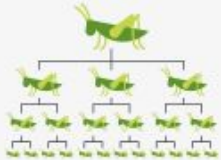
IB example:

Arctic mouse-ear chickweed is appearing earlier in spring

This may have an affect on the reindeer that migrate to graze on these plants

D4.3.11—Increases to the number of insect life cycles within a year due to climate change

HOW DOES CLIMATE CHANGE AFFECT PESTS?



Insects experience additional generations



Increased resistance to insecticides



Some insects grow bigger in warmer temperatures



Higher survival rates during winter months



Poleward spread of pests towards cooler climates



Impact of attacks on crops and people is worsened

Spruce beetle

Adult beetle burrows into tree

Lays eggs

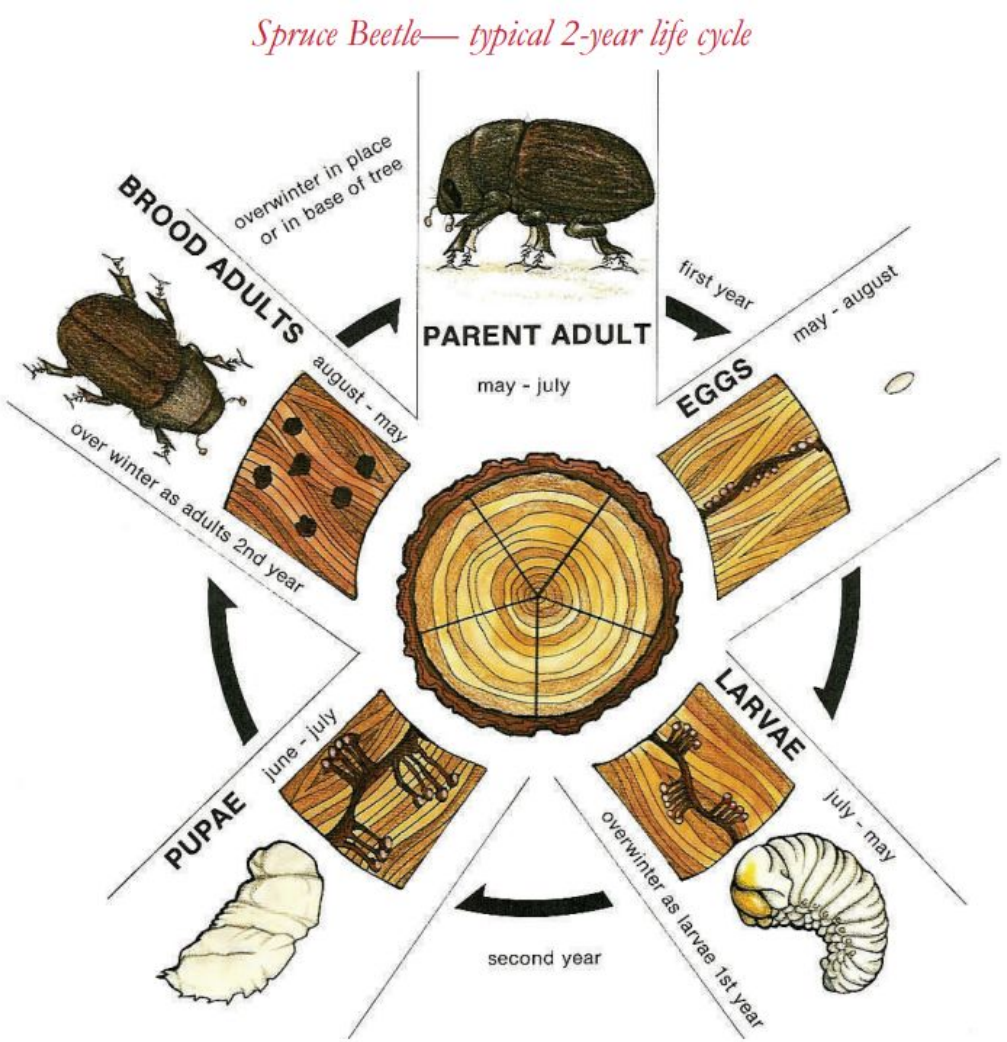
Larvae feed in phloem

Damage or kill tree



Typically has a two year life cycle.

With warmer temperatures this can increase to a one year life cycle.



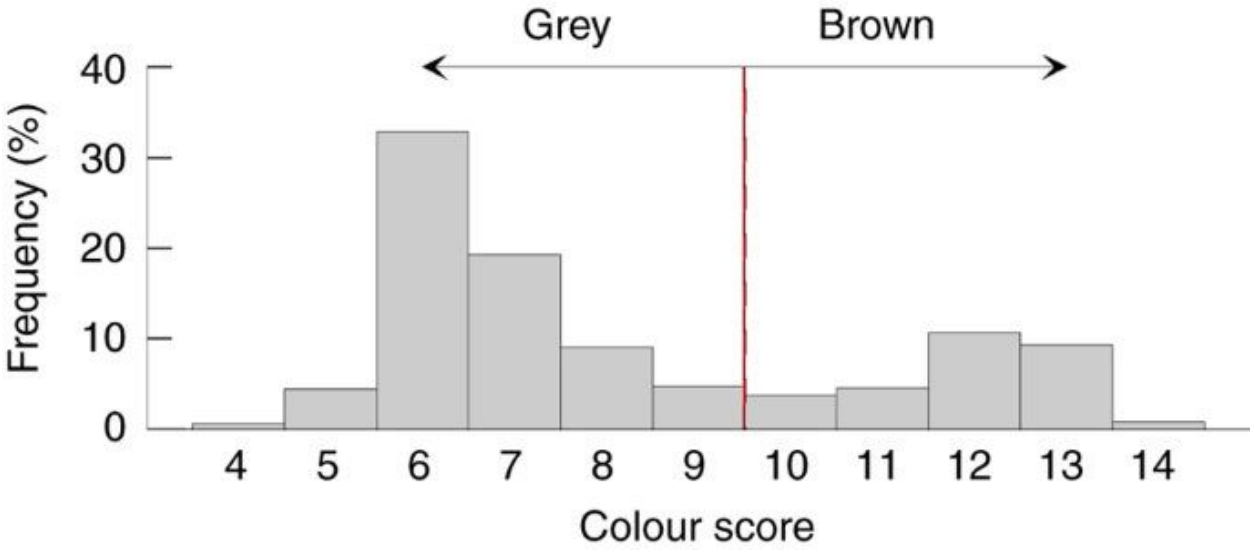
D4.3.12—Evolution as a consequence of climate change



In what ways may species evolve in response to the selection pressure of climate change?



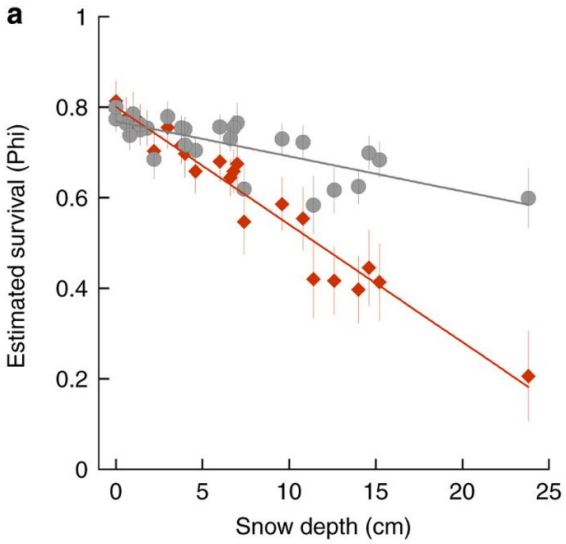


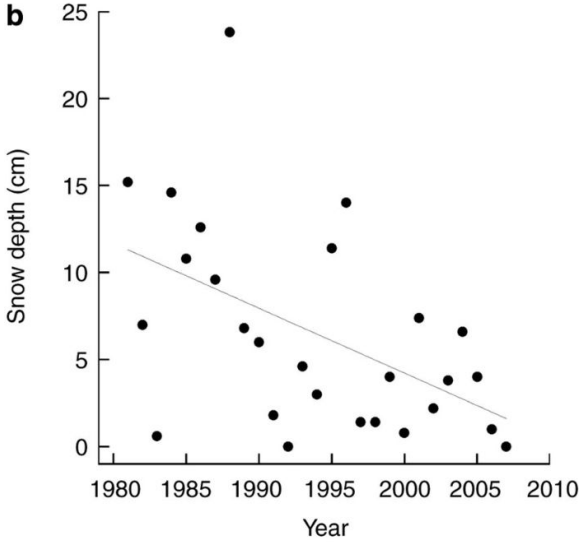

Grey
Brown
40
30
20
10
0
Frequency (%)
4
5
6
7
8
9
10
11
12
13
14
Colour score

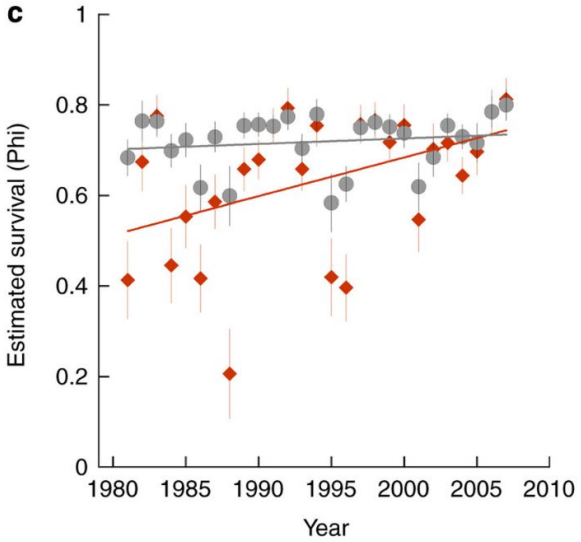
Discuss these graphs

Red diamond = brown feathers

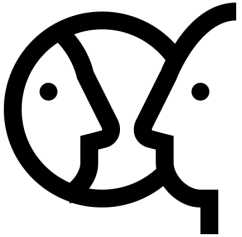
Grey circle = grey feathers







Can you...



Discuss the impact of climate change on phenological events?

Discuss how climate change can lead to evolution of species?